

CSA 0378-DATA STRUCTURES

7. Write a C program to implement Array operations such as Insert, Delete and Display

```
main.c [Icons] [Share] [Run] [Output] [Clear]
1 #include<stdio.h>
2 void main()
3 {
4     int a[20], n, i, pos, val, choice;
5
6     printf("Enter size of array: ");
7     scanf("%d", &n);
8
9     printf("Enter elements:\n");
10    for(i=0; i<n; i++)
11        scanf("%d", &a[i]);
12
13    printf("\n1.Insert 2.Delete 3.Display\nEnter choice: ");
14    scanf("%d", &choice);
15
16    switch(choice)
17    {
18        case 1:
19            printf("Enter position and value: ");
20            scanf("%d %d", &pos, &val);
21            for(i=n; i>=pos; i--)
22                a[i] = a[i-1];
23            a[pos-1] = val;
24            n++;
25            printf("After insertion: ");
26            for(i=0; i<n; i++)
27                printf("%d ", a[i]);
28            break;
```

Enter size of array: 3
Enter elements:
2
4
6

1.Insert 2.Delete 3.Display
Enter choice: 1
Enter position and value: 1
3
After insertion: 3 2 4 6

=== Code Exited With Errors ===

```
main.c [Icons] [Share] [Run] [Output] [Clear]
17 {
18     case 1:
19         printf("Enter position and value: ");
20         scanf("%d %d", &pos, &val);
21         for(i=n; i>=pos; i--)
22             a[i] = a[i-1];
23         a[pos-1] = val;
24         n++;
25         printf("After insertion: ");
26         for(i=0; i<n; i++)
27             printf("%d ", a[i]);
28         break;
29
30     case 2:
31         printf("Enter position to delete: ");
32         scanf("%d", &pos);
33         for(i=pos-1; i<n-1; i++)
34             a[i] = a[i+1];
35         n--;
36         printf("After deletion: ");
37         for(i=0; i<n; i++)
38             printf("%d ", a[i]);
39         break;
```

Enter size of array: 3
Enter elements:
2
4
6

1.Insert 2.Delete 3.Display
Enter choice: 1
Enter position and value: 1
3
After insertion: 3 2 4 6

=== Code Exited With Errors ===

```
main.c [Icons] [Share] [Run] [Output] [Clear]
27         printf("%d ", a[i]);
28         break;
29
30     case 2:
31         printf("Enter position to delete: ");
32         scanf("%d", &pos);
33         for(i=pos-1; i<n-1; i++)
34             a[i] = a[i+1];
35         n--;
36         printf("After deletion: ");
37         for(i=0; i<n; i++)
38             printf("%d ", a[i]);
39         break;
40
41     case 3:
42         printf("Array elements: ");
43         for(i=0; i<n; i++)
44             printf("%d ", a[i]);
45         break;
46 }
47 }
```

Enter size of array: 3
Enter elements:
2
4
6

1.Insert 2.Delete 3.Display
Enter choice: 2
Enter position to delete: 2
After deletion: 2 6

=== Code Exited With Errors ===